

A little-Zygmund graph with no normal: a counterexample to Conjecture 6.1 of Balashov–Repovš

Autonomous solution packet for arXiv:1007.0162

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Status: candidate full counterexample, likely valid. The construction is elementary and self-contained. Its only delicate point is a uniform second-difference estimate, proved below. No claim of priority is made: targeted cheap-index and web searches found no resolution of the conjecture, but the novelty search was not exhaustive.

1 The question

For a closed subset C of a normed space, write

$$\gamma_C(\varepsilon) = \inf \left\{ r > 0 : B_r\left(\frac{x+y}{2}\right) \cap C \neq \emptyset \text{ whenever } x, y \in C, \|x - y\| \leq \varepsilon \right\}.$$

Thus C is weakly convex on a small scale if $\gamma_C(\varepsilon) < \varepsilon/2$ there. Conjecture 6.1 of Balashov–Repovš asks, in particular, whether the assumptions

$$\gamma(\varepsilon)/\varepsilon \longrightarrow 0, \text{ and } \text{cl}(\text{int } A) = A,$$

for both A and $\text{cl}(E \setminus A)$ force the existence of a unit normal at every boundary point, uniformly continuously depending on that point. The ambient space is allowed to be uniformly convex and, parenthetically, smooth.

The answer is negative already in the Euclidean plane.

Theorem 1. *There is a globally Lipschitz function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that, for*

$$A = \{(x, y) : y \geq f(x)\}, \quad B = \text{cl}(\mathbb{R}^2 \setminus A) = \{(x, y) : y \leq f(x)\},$$

the following hold.

1. *A and B are closed and weakly convex, with $\gamma_A(\varepsilon)/\varepsilon \rightarrow 0$ and $\gamma_B(\varepsilon)/\varepsilon \rightarrow 0$.*
2. *$A = \text{cl}(\text{int } A)$.*
3. *The Fréchet normal cone of A at the origin is trivial: $N_A(0, 0) = \{0\}$.*

Consequently A has no unit normal, and no unit proximal normal, at the origin. Hence Conjecture 6.1 is false even when $E = \mathbb{R}^2$ is both uniformly convex and uniformly smooth.

2 Construction and the second-difference estimate

Fix $0 < r_0 < 1/4$ and an even function $\chi \in C_c^\infty(\mathbb{R})$ satisfying

$$\chi(x) = 1 \quad (|x| \leq r_0), \quad \chi(x) = 0 \quad (|x| \geq 2r_0).$$

Set

$$f(0) = 0, \quad f(x) = \chi(x)x \sin\left(\log \log \frac{e}{|x|}\right) \quad (x \neq 0).$$

The expression is well-defined on the support of χ . It is globally Lipschitz: near zero its derivative is bounded, and away from zero it is smooth and compactly supported.

For $h \in \mathbb{R}$, let

$$\Delta_h^2 f(x) = f(x+h) + f(x-h) - 2f(x), \quad \Omega(t) = \sup_{x \in \mathbb{R}, |h| \leq t} |\Delta_h^2 f(x)|.$$

Lemma 1 (little-Zygmund estimate). *One has $\Omega(t) = o(t)$ as $t \downarrow 0$.*

Proof. It suffices to take $h > 0$. Put

$$a(r) = \sin\left(\log \log \frac{e}{r}\right).$$

In the core $0 < |x| \leq r_0$, direct differentiation gives

$$|f''(x)| \leq \frac{C}{|x| \log(e/|x|)}. \tag{1}$$

If $|x| \geq 2h$ and $x-h, x, x+h$ remain in the core, symmetric Taylor estimates and (1) yield

$$|\Delta_h^2 f(x)| \leq C \frac{h^2}{|x| \log(e/|x|)} \leq C \frac{h}{\log(e/h)} = o(h). \tag{2}$$

On compact regions separated from zero, the same second difference is $O(h^2)$. The transition regions of χ are such compact regions.

It remains to treat $|x| < 2h$, for which all relevant arguments have absolute value at most $3h$. In the core, the second difference of the linear function $y \mapsto a(h)y$ vanishes. Therefore its absolute value is bounded by a fixed multiple of

$$hR(h), \quad R(h) = \sup_{0 \leq r \leq 3h} \frac{r}{h} |a(r) - a(h)|. \tag{3}$$

We claim $R(h) \rightarrow 0$. Given $0 < \delta < 1$, the part $r \leq \delta h$ is at most 2δ . On $\delta h \leq r \leq 3h$, write $r = ch$. Uniformly for $c \in [\delta, 3]$,

$$\log \log \frac{e}{ch} - \log \log \frac{e}{h} = \log\left(1 - \frac{\log c}{\log(e/h)}\right) \rightarrow 0.$$

Since sine is Lipschitz, the remaining part of (3) tends to zero. First choose δ small and then h small. Thus (3) is $o(h)$.

Combining this with (2), and taking the monotone supremum over $|h| \leq t$, proves $\Omega(t) = o(t)$. $\square$

3 Weak convexity of both sides of the graph

Lemma 2. *For every $\varepsilon > 0$,*

$$\gamma_A(\varepsilon), \gamma_B(\varepsilon) \leq \frac{1}{2}\Omega(\varepsilon/2).$$

In particular both exact moduli are $o(\varepsilon)$.

Proof. Take $z_i = (x_i, y_i) \in A$ with $\|z_1 - z_2\| \leq \varepsilon$ and set $x_m = (x_1 + x_2)/2$, $y_m = (y_1 + y_2)/2$. Since $y_i \geq f(x_i)$,

$$f(x_m) - y_m \leq f(x_m) - \frac{f(x_1) + f(x_2)}{2} \leq \frac{1}{2} |\Delta_{(x_1-x_2)/2}^2 f(x_m)|.$$

If the midpoint is outside A , moving it vertically upward by the positive quantity on the left reaches A . Its distance to A is therefore at most $\Omega(\varepsilon/2)/2$. This proves the estimate for A .

For B , reverse the inequalities and move a midpoint vertically downward. The same absolute second difference gives the identical bound. Lemma 1 now gives both $o(\varepsilon)$ assertions. In particular the displayed common upper majorant is less than $\varepsilon/2$ for all sufficiently small ε , so both sets are weakly convex. $\square$

Closedness and the identity $A = \text{cl}(\text{int } A)$ follow immediately from the continuity of f .

4 Failure of every nonzero normal

The Fréchet normal cone at $z \in A$ is

$$N_A(z) = \{p : \langle p, w - z \rangle \leq o(\|w - z\|) \text{ as } w \rightarrow z, w \in A\}.$$

Suppose $p = (u, v) \in N_A(0, 0)$. Testing on the vertical points $(0, s)$, $s > 0$, gives $v \leq 0$.

If $v = 0$, testing on boundary points $(t, f(t))$ with both signs of t forces $u = 0$. Suppose instead that $v < 0$, and put $q = -u/v$. On boundary points, after dividing by the positive number $-v$, the normal inequality implies

$$f(t) - qt \geq -o(|t|). \tag{4}$$

For $t > 0$, the quotient

$$\frac{f(t)}{t} = \sin\left(\log \log \frac{e}{t}\right)$$

has subsequences converging to both -1 and 1 . Applying (4) along a subsequence converging to -1 gives $q \leq -1$. For $t < 0$, division of (4) by the negative number t reverses the inequality; along a subsequence for which $f(t)/t \rightarrow 1$, it gives $q \geq 1$. This is impossible. Consequently $p = 0$, so $N_A(0, 0) = \{0\}$.

Every proximal normal in a Hilbert space is a Fréchet normal. Thus there is also no nonzero proximal normal at the origin, completing the proof of Theorem 1.

5 Interpretation and verification

The construction separates the two regularity scales at the heart of the conjecture. The estimate $\Delta_h^2 f = o(h)$ makes both sides of the graph asymptotically convex in precisely the required midpoint sense, while the first-order slopes still oscillate through the whole interval $[-1, 1]$. The hypotheses therefore control symmetric second differences but do not force a supporting first-order direction.

The accompanying script `code/verify_second_difference.py` samples the normalized second differences and the oscillating secants. It is only a numerical sanity check; the proof is Lemmas 1 and 2.

Remark 1. *The source paper's stronger theorem assumes a quadratic bound on the modulus. Here the decay is only of order comparable to $\varepsilon/\log(e/\varepsilon)$ along the difficult scales. Hence the example does not contradict that theorem; it shows why the conjectured replacement by the qualitative condition $o(\varepsilon)$ is too weak.*

Source context

The crop below is from page 13 of the locally rendered source arXiv:1007.0162. It includes the quadratic theorem, the authors' epilogue, and Conjecture 6.1.

Theorem 5.2. *Let space E be uniformly convex with modulus of convexity of the second order and uniformly smooth, and let subsets $A \subset E$ and $\text{cl}(E \setminus A)$ be weakly convex with modulus $\gamma(\varepsilon)$ of the second order and $\text{cl int } A = A$. Then $N(A, x) \cap \partial B_1^*(0) = \{p(x)\}$, $N(\text{cl}(E \setminus A), x) \cap \partial B_1^*(0) = \{-p(x)\}$, at any point $x \in \partial A$ and $p(x)$ uniformly continuously depends on the point $x \in \partial A$.*

Proof. By Remark 2.2 the sets A and $\text{cl}(E \setminus A)$ are proximally smooth with some parameter $d > 0$. Using the results of Ivanov [11, Theorem 2] we have that for any point $x \in \partial A$ there exists a proximally normal vector $n(x)$ to the set A at the point $x \in \partial A$ (and proximally normal vector $-n(x)$ to the set $\text{cl}(E \setminus A)$ at the point $x \in \partial A$) and $n(x)$ uniformly continuously depends on x . By uniform smoothness of the space E for any vector $n \in E$, $\|n\| = 1$, there exists unit vector $p(n) \in E^*$ with $(p(n), n) = 1$ and $p(n)$ uniformly continuously depends on n ([9, 14]). Again by the smoothness of the space E we have that $p(x) = p(n(x)) \in N(A, x)$ is uniformly continuous on $x \in \partial A$. $\square$

6. EPILOGUE

1. We see from the results above that in the spaces with modulus of convexity of the second order the notion of weakly convex set is very effective.

2. Some of the results can be proved in a more general setting of uniformly convex Banach spaces: Theorem 3.4, or Theorems 3.1, 3.2, and 3.3 (see [2] for details). However, the proofs in [2] are much more complicated.

We wish to point out that the results above are interesting and nontrivial even in the finite-dimensional case.

3. We agree with Banaś that the modulus σ from [4] and [5] is sometimes much more convenient in applications than the standard modulus of smoothness [9], [14]. In fact, the modulus of nonconvexity is a modification of modulus σ from [4] for the nonconvex case.

Also, Theorem 5.1 shows a deep relationship between weakly convex sets and smooth sets. In conclusion, we formulate the following

Conjecture 6.1. *Let the space E be uniformly convex (and smooth?), the subsets $A \subset E$ and $\text{cl}(E \setminus A)$ closed and weakly convex with modulus $\gamma(\varepsilon)$ and $\text{cl int } A = A$. If $\lim_{\varepsilon \rightarrow +0} \gamma(\varepsilon)/\varepsilon = 0$, then the unit normal vector to the set A at the point $x \in \partial A$ uniformly continuously depends on the point $x \in \partial A$.*

References

- [1] M. V. Balashov and D. Repovš, *Weakly convex sets and modulus of nonconvexity*, arXiv:1007.0162.